# Supplementary Material for LFTNet

This supplementary material provides additional visual analyses for the proposed LFTNet, including Local Attribution Map (LAM) analysis and qualitative comparisons on challenging image regions. These results complement the quantitative evaluations reported in the main paper.

## 1 Local Attribution Map Analysis

We further investigate the effective receptive field of different super-resolution models using Local Attribution Maps (LAM). LAM measures the contribution of input pixels to the reconstruction of a selected output region, providing an intuitive visualization of the spatial information utilized by each model.

As shown in Fig. 1, conventional convolution-based approaches exhibit relatively concentrated attribution regions, indicating that their reconstruction mainly relies on local neighborhood information. In contrast, attention-based methods generally utilize broader spatial regions.

LFTNet exhibits the widest and more distributed attribution pattern among the compared methods. Quantitatively,LFTNet achieves a Diffusion Index (DI) of 21.90, compared with 4.07 for EDSR [1], 7.61 for SwinIR [2], 11.51 for SRFormer [3], and 20.89 for MambaIRv2 [4]. A higher DI indicates that information from a larger spatial region contributes to the reconstruction.

These results provide additional evidence that the proposed LMSA effectively enlarges the receptive field and facilitates long-range information interaction, which is particularly beneficial for reconstructing complex structures and repetitive textures.

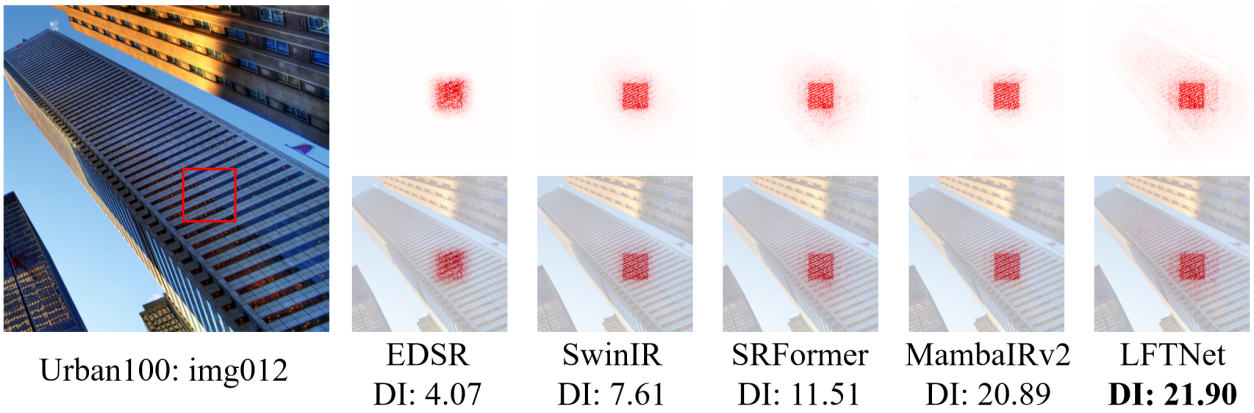



Figure 1: Local Attribution Map (LAM) comparison among different super-resolution methods. The Diffusion Index (DI) measures the spatial extent of input information involved in reconstruction.

## 2 Qualitative Comparison

Fig. 2 presents additional qualitative comparisons on challenging image regions containing repetitive textures, building structures, and dense line patterns.

For building facades and stripe-like structures, bicubic interpolation and several lightweight SR methods tend to produce over-smoothed textures, distorted edges, or inconsistent orientations. In comparison, LFTNet better preserves edge continuity and directional consistency, resulting in clearer and more regular structural patterns.

For dense grids and interleaved line patterns, competing approaches may produce broken structures, blurred boundaries, or adhesion artifacts between adjacent lines. LFTNet reconstructs these fine structures more reliably and maintains more regular spacing and geometric consistency, producing results that are closer to the ground-truth HR images.

These visual comparisons further demonstrate the effectiveness of combining long-range spatial modeling with frequency-domain feature modulation for recovering structured and high-frequency image details.

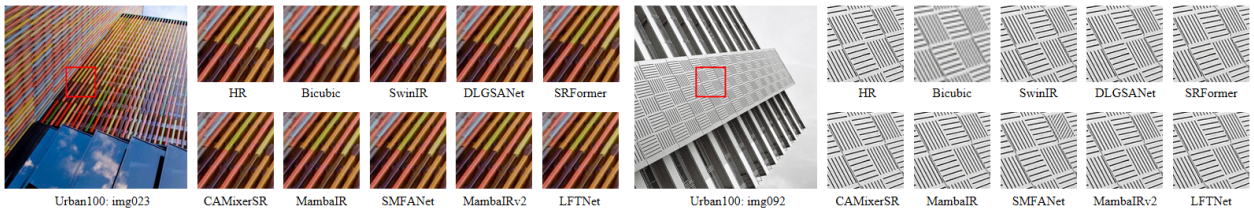



Figure 2: Additional qualitative comparisons of lightweight image super-resolution methods on challenging image regions.

## 3 Discussion

The LAM and qualitative results provide complementary evidence to the quantitative evaluations in the main paper. LMSA enables the model to exploit a broader spatial context through direction-complementary long-range interaction, while FGFN enhances informative structural and texture components through frequency-selective modulation. Their combination improves the reconstruction of both large-scale structures and fine repetitive details under a lightweight model budget.

## References

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